

Architecture Changes

Crypto as-is vs target, and the proposed Indian equity engine. 8 Oct 2026.

Current crypto architecture vs the target, then the proposed architecture for the new Indian equity engine.
The theme: separate the always-on collector from the trading decision path, and make the ledger the single source of truth.

1. Current crypto architecture (as-is)

```
EC2 (single box, 17d up)                Aiven free Postgres (54MB)
  crypto-engine.service                  crypto_snapshots 32MB/94.9k rows
    collectors -> DB -> analysis -> book
    health.py (8,131 lines - god file)
    deploy: main -> GitHub Actions -> EC2 (test gate)
ThinkPad dhruv-ai: lab backtests + Ollama AI gate
Mac: dash.py monitor :8765 + lab_pull + watchdog
```

2. Pain points found in the audit

- Single point of failure: 1 EC2 + free-tier DB, no failover.
- health.py is an 8,131-line god file - hard to change safely.
- No hard position-reconciliation loop (C1 orphan-position bug proved it).
- DB TLS unverified (CERT_NONE) - creds exposed to MITM.
- In-sample constants baked into the live path.
- execution/ never placed a real order (fake-exchange tested only).

3. Target architecture (service-separated)

```
[collector svc] --bars/news--> [Postgres (TLS, backed up)]
                                   |
[decision svc] <--signals-- [analysis/lab]
                                   | risk caps + reconciliation
[execution svc] --orders--> [exchange/broker]
                                   |
[ledger svc] (append-only, source of truth) -> [dashboard]
```

Service	Level	Why
Collector	always-on, stateless	just ingests bars/news; restart freely
Analysis / lab	batch, offline	backtests + stats; never in the live path
Decision	live, guarded	applies risk caps; the only writer of intents
Execution	live, minimal	turns intents into orders; reconciles positions
Ledger	append-only	single source of truth; every service reads it
Dashboard	read-only	observability; survives restarts (watchdog)

4. New Indian equity engine - proposed architecture

```
Kite Connect (Zerodha)
  | historical daily bars + live quotes (09:15-15:30 IST)
  v
[data lake: NSE daily bars, Nifty-50] -> [lab: backtest + stats]
                                   |
[paper engine: market-hours book] <--- signals ---
  | risk caps (0.5-1%), reconciliation, kill switch
  v
```

```
[ledger: append-only] -> [dashboard + watchdog]
|
[execution svc] --(PAPER by default; real only after gate)--> Kite orders
```

5. What changes vs crypto

- Market-hours, not 24/7: the engine wakes 09:00-15:30 IST and idles otherwise.
- No funding-rate mechanics; cost model is STT/stamp/GST/DP/brokerage instead.
- Daily bars are the swing timeframe (4h bars are awkward inside a 6h15m session).
- No overnight short in cash equity; intraday auto-square-off ~15:20 must be handled.
- Reconciliation and the ledger are built in from day 1 (the crypto C1 bug lesson).
- DB TLS verified from the start; secrets in a store, never loose at repo root.